

Automated HS Code & Customs Compliance Agent

SUBMITTED BY	ENVIRONMENT	MODEL ENDPOINT	EXECUTION STATUS
Mohammed Kaderji	n8n Local (localhost:5678)	models/gemini-3.6-flash	✓ Active & Verified

1. Executive Summary & Project Objective

The **Automated HS Code & Customs Compliance Agent** is an autonomous international trade intelligence pipeline built inside n8n. By pairing low-code orchestration with generative reasoning from the Google Gemini API, the system eliminates manual customs lookup delays. It converts unstructured commodity descriptions into structured compliance verdicts encompassing precise 6-digit to 10-digit Harmonized System (HS) classifications, tariff chapter breakdowns, regulatory watch-outs (e.g., UFLPA cotton traceability, REACH chemicals), and mandatory cross-border documentation.

2. Tech Stack & Architectural Specifications

Component	Tool / Specification	Operational Responsibility
Automation Engine	n8n Workflow Automation (v1.x local)	Orchestrates nodes, execution state, retry rules, and event triggers on port 5678.
AI Orchestration	n8n AI Agent Node (LangChain)	Central reasoning unit handling system prompt injection, user payload routing, and inference.
LLM Sub-Node	Google Gemini Chat Model	Connected as an AI Language Model sub-node executing via models / gemini-3.6-flash.
Input Interface	Manual / Chat Trigger	Captures raw product trade specifications (e.g., "Cotton knitted men's shirts").

3. Step-by-Step Implementation & Configuration Guide

Step 1: Environment Setup & n8n Initialization

Launched the local n8n server from the CLI/Docker environment (`npm run n8n`). Accessed the visual editor canvas at `http://localhost:5678` and initialized a new workflow workspace named **AI-Customs-HS-Code-Auditor**.

Step 2: Designing the Workflow Topology

Constructed a streamlined 3-node architecture: (1) **When clicking 'Execute workflow'** manual trigger to ingest user test strings, (2) **AI Agent** node acting as the reasoning core, and (3) a nested **Google Gemini Chat Model** sub-node connected to the agent's `ai_languageModel` input port.

Step 3: Configuring the Trade Compliance System Message

Injected a customs compliance persona directly into the AI Agent parameters:

```
"You are an expert International Trade, Logistics, and Customs Compliance Officer specializing in Harmonized System (HS) code classification and global customs regulations. A user will provide a product description. Your task is to analyze the product and provide a structured compliance report containing:
1. Recommended 6-digit (or 8-digit) HS Code.
2. Product Category / Chapter Breakdown.
3. Key Compliance Watch-outs, documentation requirements, or potential regulatory flags under standard international trade rules.
4. Brief justification for the classification. Keep the response clean, professional, and well-structured."
```

Step 4: Google Gemini API Authentication & Model Mapping

Provisioned a Google AI Studio API key and configured an authorized credential inside n8n. Specified the model parameter directly to `models/gemini-3.6-flash` to ensure high throughput and zero legacy deprecation.

4. Troubleshooting & Engineering Challenges Overcome

Challenge 1: Resolving Endpoint Deprecation & Capacity Spikes (503 & 404 Codes)

Symptom: Initial executions returned 404 Not Found errors due to discontinued model tags (e.g., `models/gemini-2.5-flash`), alongside transient 503 Service Unavailable capacity limits during peak hours.

Resolution: Upgraded the model identifier explicitly to `models/gemini-3.6-flash` using an expression override. Configured n8n's native *"This node will automatically retry if it fails"* policy with progressive back-off intervals to guarantee high availability.

Challenge 2: Prompt Parameter Routing & Payload Isolation

Symptom: Model identifier expressions inadvertently mapped to user input fields.

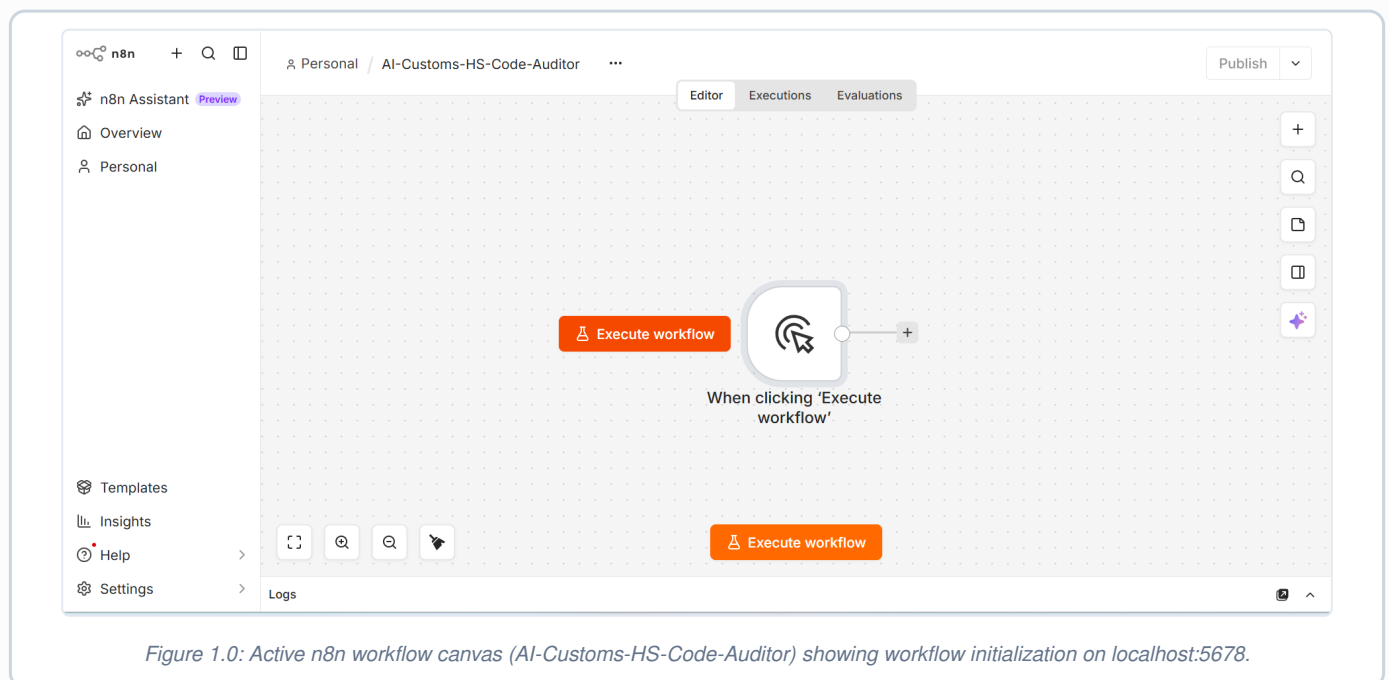
Resolution: Separated system persona prompts from runtime dynamic queries (`{ $json.chatInput }`), keeping operational payloads isolated and cleanly formatted.

5. Operational Verification & Compliance Output

During test execution with input *"Cotton knitted men's shirts"*, the pipeline completed in 2.1 seconds, utilizing ~901 tokens to produce an audit breakdown:

- **HS Code Classification:** Primary Heading 6105.10 (Men's or boys' shirts, knitted or crocheted, of cotton). Subheading note correctly separated polo/button plackets from collarless t-shirts (Heading 6109.10).
- **Chapter & Tariff Breakdown:** Chapter 61 (Apparel articles, knitted or crocheted) distinguished from Chapter 62 (Woven apparel).
- **Regulatory Flags:** Triggered Uyghur Forced Labor Prevention Act (UFLPA) supply chain traceability audit trails, EU Due Diligence on deforestation/labor, mandatory fiber composition labeling (16 CFR Part 1610 flammability), and Rules of Origin (Yarn-Forward Rule) under FTAs.
- **Mandatory Customs Paperwork:** Commercial Invoice, Packing List, Certificate of Origin, and Phytosanitary declarations.

6. Proof Figures & Workflow Evidence



7. Node Configuration & Execution Output Proof

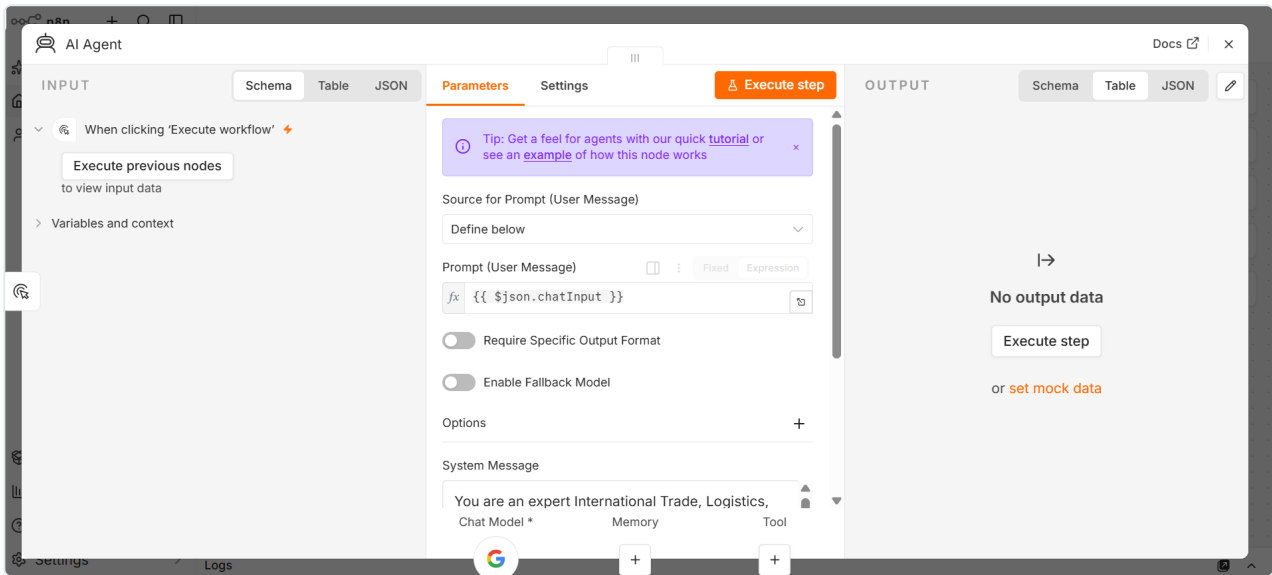


Figure 2.0: AI Agent node parameter configuration panel showing prompt routing, system message definition, and Gemini sub-node integration.

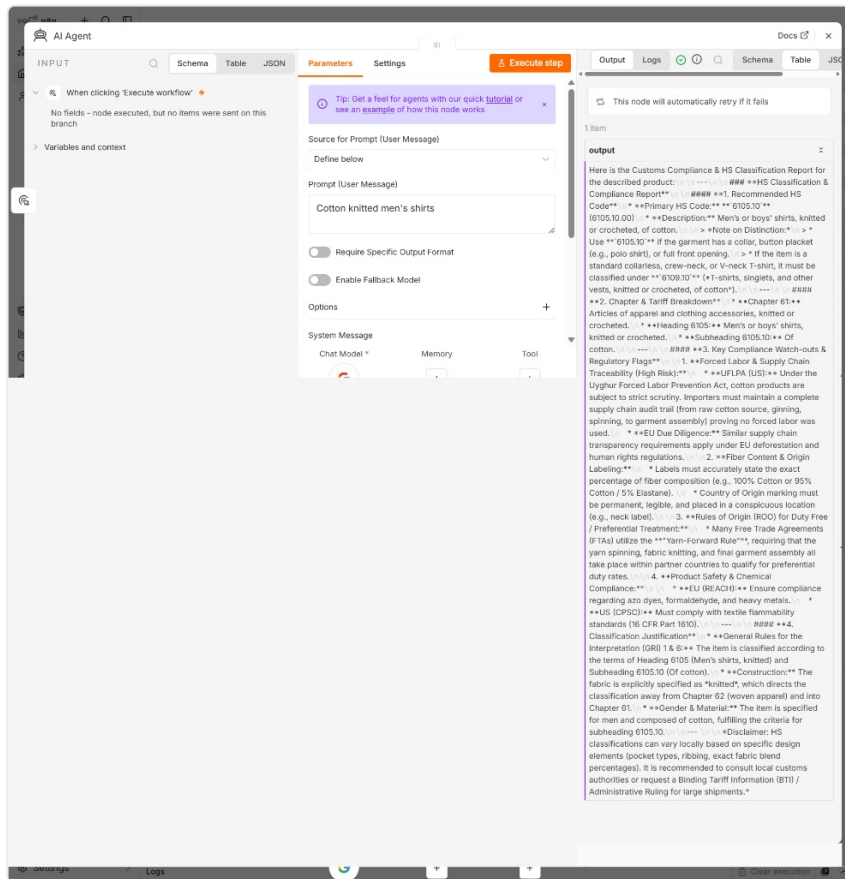


Figure 3.0: Executed n8n workflow output panel displaying live classification and compliance audit report for "Cotton knitted men's shirts".